50-30-20 Budget - Dev

California CCSS Mathematics Standards Alignment

OBJECTIVES	CODE ASSIGNMENTS	HIGH CONFIDENCE	GRADE BAND
27	**27**	**12**	**9-12**
15 aligned · 12 not	12 distinct standards	10 med · 5 low	inferred from context

Personal-finance spreadsheet project at HS Algebra/Modeling level. Most concrete math is in the formula objectives (net pay, percentages, sums) which map cleanly to A-SSE / N-Q / 7.RP. Several objectives are pure data entry, categorization, or reflective writing and have no math alignment. The pie-chart task lacks a clean CCSS-M anchor - MP.4 (modeling) is the strongest available fit.

Section 1: Project Setup

Watch Me

Watch the introductory video.

`sections[0].objectives[0]`

Note: Platform onboarding. No math standard applies.

Platform onboarding. No math standard applies.

Personal Information

Use AI to research careers and fill in personal info.

`sections[0].objectives[1]`

Note: Career research / data entry. No math standard applies (closest ELA alignment is W.9-10.7/8 research).

Career research / data entry. No math standard applies (closest ELA alignment is W.9-10.7/8 research).

Tax and Income

Use AI to research effective tax rate and income for a chosen career.

`sections[0].objectives[2]`

N-Q.A.1 N | Grade 9-12 | *Reason quantitatively and use units to solve problems.* | **90** / 100 **HIGH**

Use units as a way to understand problems and to guide the solution of multi-step problems; choose and interpret units consistently in formulas; choose and interpret the scale and the origin in graphs and data displays. ■

WHY THIS MAPS

Use units to understand problems and choose units consistently - annual gross income (USD) and effective tax rate (%) must be reported with appropriate units.

7.RP.A.3 RP | Grade 7 | *Analyze proportional relationships and use them to solve real-world and mathematical problems.* | **70** / 100 **MEDIUM**

Use proportional relationships to solve multistep ratio and percent problems. Examples: simple interest, tax, markups and markdowns, gratuities and commissions, fees, percent increase and decrease, percent error.

WHY THIS MAPS

Effective tax rate is a percent applied to income. Even at HS, the underlying ratio reasoning is the 7.RP percent-of-quantity skill.

Section 2: Formulas

Annual Net Pay

Write a formula: [Annual Gross Income] * (1 - [Effective Tax Rate]).

`sections[1].objectives[0]`

A-SSE.A.1.b A | Grade 9-12 | *Interpret the structure of expressions.* | **90** / 100 **HIGH**

Interpret complicated expressions by viewing one or more of their parts as a single entity. For example, interpret P(1+r) n as the product of P and a factor not depending on P. ■

WHY THIS MAPS

Interpret complicated expressions by viewing parts as a single entity - (1 - tax_rate) is the after-tax fraction acting as a single multiplier on gross income.

A-SSE.A.1.a A | Grade 9-12 | *Interpret the structure of expressions.* | **70** / 100 **MEDIUM**

Interpret parts of an expression, such as terms, factors, and coefficients. ■

WHY THIS MAPS

Interpret parts of an expression: identifying gross income, tax rate, and the subtraction-from-one structure.

Monthly Net Pay

Write a formula: annual net pay / 12.

`sections[1].objectives[1]`

A-SSE.A.1.b A | Grade 9-12 | *Interpret the structure of expressions.* | **90** / 100 **HIGH**

Interpret complicated expressions by viewing one or more of their parts as a single entity. For example, interpret P(1+r) n as the product of P and a factor not depending on P. ■

WHY THIS MAPS

Same compound-expression interpretation; annual_net / 12 reuses the annual-net result as a single quantity.

N-Q.A.1 N | Grade 9-12 | *Reason quantitatively and use units to solve problems.* | **70** / 100 **MEDIUM**

Use units as a way to understand problems and to guide the solution of multi-step problems; choose and interpret units consistently in formulas; choose and interpret the scale and the origin in graphs and data displays. ■

WHY THIS MAPS

Unit conversion (USD/year -> USD/month) - choosing and tracking units is the core skill here.

Section 3: Needs

Housing

Enter housing type and monthly cost.

`sections[2].objectives[0]`

Note: Data entry. No math standard applies.

Data entry. No math standard applies.

Needs List

List at least seven needs with notes explaining each.

`sections[2].objectives[1]`

Note: Data entry and categorization with brief written notes. No math standard applies.

Data entry and categorization with brief written notes. No math standard applies.

Sum Function

Use SUM() to total monthly need costs.

`sections[2].objectives[2]`

F-IF.A.1 F | Grade 9-12 | *Understand the concept of a function and use function notation.* | **90** / 100 **HIGH**

Understand that a function from one set (called the domain) to another set (called the range) assigns to each element of the domain exactly one element of the range. If f is a function and x is an element of its domain, then f(x) denotes the output of f corresponding to the input x. The graph of f is the graph of the equation y = f(x).

WHY THIS MAPS

SUM() is a function from a domain (range of cells) to a single output value - exactly the function-mapping definition.

F-IF.A.2 F | Grade 9-12 | *Understand the concept of a function and use function notation.* | **70** / 100 **MEDIUM**

Use function notation, evaluate functions for inputs in their domains, and interpret statements that use function notation in terms of a context.

WHY THIS MAPS

Use function notation and evaluate functions for inputs in their domain - SUM(C12:C26) uses spreadsheet function notation in context.

% Monthly Net Pay

Calculate percentage of monthly net pay spent on needs.

`sections[2].objectives[3]`

7.RP.A.3 RP | Grade 7 | *Analyze proportional relationships and use them to solve real-world and mathematical problems.* | **90** / 100 **HIGH**

Use proportional relationships to solve multistep ratio and percent problems. Examples: simple interest, tax, markups and markdowns, gratuities and commissions, fees, percent increase and decrease, percent error.

WHY THIS MAPS

Percent-of-quantity problem - needs subtotal as a percentage of monthly net pay is exactly this 7.RP skill, even when used at HS.

N-Q.A.1 N | Grade 9-12 | *Reason quantitatively and use units to solve problems.* | **70** / 100 **MEDIUM**

Use units as a way to understand problems and to guide the solution of multi-step problems; choose and interpret units consistently in formulas; choose and interpret the scale and the origin in graphs and data displays. ■

WHY THIS MAPS

Choosing units (decimal vs %) and formatting the cell as percentage.

Section 4: Wants

Entertainment

Enter entertainment line-item cost.

`sections[3].objectives[0]`

Note: Data entry. No math standard applies.

Data entry. No math standard applies.

Wants List

List at least seven wants with notes.

`sections[3].objectives[1]`

Note: Data entry and categorization. No math standard applies.

Data entry and categorization. No math standard applies.

Reusing Formulas

Copy and paste cell formulas to reuse the SUM() pattern.

`sections[3].objectives[2]`

MP.7	MP	Grade —	*Mathematical Practice*	**90** / 100 **HIGH**

Look for and make use of structure.

WHY THIS MAPS

Look for and make use of structure - recognizing the SUM-of-range pattern transfers across the Wants subtotal.

MP.8	MP	Grade —	*Mathematical Practice*	**70** / 100 **MEDIUM**

Look for and express regularity in repeated reasoning.

WHY THIS MAPS

Express regularity in repeated reasoning - same calculation structure applied to a different data range.

% Monthly Net Pay (Wants)

Calculate % of net pay spent on wants.

`sections[3].objectives[3]`

7.RP.A.3	RP	Grade 7	*Analyze proportional relationships and use them to solve real-world and mathematical problems.*	**90** / 100 **HIGH**

Use proportional relationships to solve multistep ratio and percent problems. Examples: simple interest, tax, markups and markdowns, gratuities and commissions, fees, percent increase and decrease, percent error.

WHY THIS MAPS

Percent-of-quantity, identical to the Needs % task.

N-Q.A.1	N	Grade 9-12	*Reason quantitatively and use units to solve problems.*	**70** / 100 **MEDIUM**

Use units as a way to understand problems and to guide the solution of multi-step problems; choose and interpret units consistently in formulas; choose and interpret the scale and the origin in graphs and data displays. ■

WHY THIS MAPS

Tracking units and formatting as percent.

Section 5: Financials

Emergency Fund

Enter monthly emergency fund contribution.

`sections[4].objectives[0]`

Note: Data entry. No math standard applies.

Data entry. No math standard applies.

Financials List

List at least four financial goals with monthly contributions.

`sections[4].objectives[1]`

Note: Data entry and categorization. No math standard applies.

Data entry and categorization. No math standard applies.

Modify Copied Formulas

Copy, paste, and adjust cell formulas for the Financial Goals subtotal.

`sections[4].objectives[2]`

MP.7 MP | Grade — | *Mathematical Practice* | **90** / 100 **HIGH**

Look for and make use of structure.

WHY THIS MAPS

Look for and make use of structure - adapting the same SUM/% pattern with adjusted ranges and absolute references.

A-SSE.A.1.b A | Grade 9-12 | *Interpret the structure of expressions.* | **50** / 100 **LOW**

Interpret complicated expressions by viewing one or more of their parts as a single entity. For example, interpret P(1+r) n as the product of P and a factor not depending on P. ■

WHY THIS MAPS

Treating the subtotal expression as a single quantity that the percent formula divides into monthly net pay.

% Monthly Net Pay (Financials)

Calculate % of net pay spent on financial goals.

`sections[4].objectives[3]`

7.RP.A.3 RP | Grade 7 | *Analyze proportional relationships and use them to solve real-world and mathematical problems.* | **90** / 100 **HIGH**

Use proportional relationships to solve multistep ratio and percent problems. Examples: simple interest, tax, markups and markdowns, gratuities and commissions, fees, percent increase and decrease, percent error.

WHY THIS MAPS

Percent-of-quantity, same skill as the Needs and Wants % tasks.

Section 6: Totals and Chart

Total Spending

Formula adding the three subtotals.

`sections[5].objectives[0]`

A-SSE.A.1.b A | Grade 9-12 | *Interpret the structure of expressions.* | **90** / 100 **HIGH**

Interpret complicated expressions by viewing one or more of their parts as a single entity. For example, interpret P(1+r) n as the product of P and a factor not depending on P. ■

WHY THIS MAPS

Interpret a sum of subtotals as a compound expression - each subtotal is itself a sum, and the total is a sum-of-sums.

F-IF.A.1 F | Grade 9-12 | *Understand the concept of a function and use function notation.* | **50** / 100 **LOW**

Understand that a function from one set (called the domain) to another set (called the range) assigns to each element of the domain exactly one element of the range. If f is a function and x is an element of its domain, then f(x) denotes the output of f corresponding to the input x. The graph of f is the graph of the equation y = f(x).

WHY THIS MAPS

Spreadsheet sum is a function over the three subtotal inputs producing one output.

Remaining Funds

Formula subtracting total spending from net pay; percent formulas for E55/E56.

`sections[5].objectives[1]`

A-SSE.A.1.b A | Grade 9-12 | *Interpret the structure of expressions.* | **90** / 100 **HIGH**

Interpret complicated expressions by viewing one or more of their parts as a single entity. For example, interpret P(1+r) n as the product of P and a factor not depending on P. ■

WHY THIS MAPS

[Net Pay] - [Total Spending] reads each operand as a single quantity; same compound-expression interpretation.

7.RP.A.3 RP | Grade 7 | *Analyze proportional relationships and use them to solve real-world and mathematical problems.* | **70** / 100 **MEDIUM**

Use proportional relationships to solve multistep ratio and percent problems. Examples: simple interest, tax, markups and markdowns, gratuities and commissions, fees, percent increase and decrease, percent error.

WHY THIS MAPS

The accompanying percent formulas in E55/E56 are percent-of-quantity calculations.

Pie Chart

Create a pie chart visualizing the budget breakdown.

`sections[5].objectives[2]`

Note: CCSS-M does not directly anchor pie charts at HS; closest content standard would be middle-grades data display (6.SP), but Mathematical Practices are a more honest fit.

MP.4 MP Grade — *Mathematical Practice* **70** / 100 **MEDIUM**

Model with mathematics.

WHY THIS MAPS

Model with mathematics - choosing a pie chart to represent categorical proportions of a budget is a modeling choice.

MP.5 MP Grade — *Mathematical Practice* **50** / 100 **LOW**

Use appropriate tools strategically.

WHY THIS MAPS

Use appropriate tools strategically - selecting and configuring a chart tool for the data.

Section 7: Reflections and Notes

Notes

Write notes explaining at least eight budget items.

`sections[6].objectives[0]`

Note: Primarily an ELA writing task; minimal math content beyond precision-of-quantity.

MP.6 MP Grade — *Mathematical Practice* **50** / 100 **LOW**

Attend to precision.

WHY THIS MAPS

Attend to precision - explanatory notes that pin down specifics (year/make/model, sqft, etc.) tighten the model. Primary alignment is ELA writing (W.9-10.4).

Spreadsheet Reflection

Write a 25+ word reflection on spreadsheet skills learned.

`sections[6].objectives[1]`

Note: Reflective/metacognitive writing. No math standard applies.

Reflective/metacognitive writing. No math standard applies.

50/30/20 Reflection

Write a 25+ word reflection on the 50/30/20 budgeting rule.

`sections[6].objectives[2]`

Note: Reflective/metacognitive writing. No math standard applies.

Reflective/metacognitive writing. No math standard applies.

AI Reflection

Write a 25+ word reflection on using AI in this project.

`sections[6].objectives[3]`

Note: Reflective/metacognitive writing. No math standard applies.

Reflective/metacognitive writing. No math standard applies.

Section 8: Teacher Graded

Budget Realism (Teacher Graded)

Check the budget tells a believable financial story.

`sections[7].objectives[0]`

F-LE.B.5 F Grade 9-12 *Interpret expressions for functions in terms of the situation they model.* **90** / 100 **HIGH**

Interpret the parameters in a linear or exponential function in terms of a context. ■

WHY THIS MAPS

Interpret parameters in a function in terms of context - a 'realistic' budget is precisely about whether the parameters (incomes, costs, percentages) are sensible in context.

MP.4 MP Grade — *Mathematical Practice* **70** / 100 **MEDIUM**

Model with mathematics.

WHY THIS MAPS

Model with mathematics - judging whether the model's assumptions and outputs are reasonable.

Reflection Quality (Teacher Graded)

Grade the three reflections for depth and specificity.

`sections[7].objectives[1]`

Note: Rubric for reflective writing. No math standard applies (closest ELA alignment is W.9-10.4).

Rubric for reflective writing. No math standard applies (closest ELA alignment is W.9-10.4).

Accuracy and Completeness (Teacher Graded)

Check items are placed in the right category (Needs/Wants/Financials).

`sections[7].objectives[2]`

Note: Categorization rubric. Primary discipline is financial literacy / CTE rather than CCSS-M.

MP.6 MP Grade — *Mathematical Practice* **50** / 100 **LOW**

Attend to precision.

WHY THIS MAPS

Attend to precision - correct categorization is a precision-of-classification check, but this is mostly a financial-literacy / CTE rubric.